EXACT CONTROLLABILITY FOR STOCHASTIC FIRST-ORDER MULTI-DIMENSIONAL HYPERBOLIC SYSTEMS*

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Abstract. We study exact controllability for stochastic first-order symmetric hyperbolic systems in several space dimensions. The principal matrices are smooth and need not commute. On smooth domains we impose a two-sided nested spectral condition on the incoming and outgoing boundary spaces, allowing the boundary matrix to change rank. Boundary inputs are measured in the natural incoming flux norm. Under a matrix escape condition, a boundary drift control and a distributed control acting in the diffusion yield exact controllability to every square-integrable terminal random state whenever the control time exceeds the oscillation of the escape weight divided by its coercivity constant. The proof combines the classical weak–strong theory for nonuniformly characteristic boundaries with exact adjoint-domain identification, a stochastic flux-trace construction, and a weighted observability estimate. Noncommuting shallow-water and acoustic–convective systems, including variable-coefficient examples, are covered. Complementary noncontrollability results show the necessity of the control configuration under the stated additional hypotheses.

Key words. Stochastic symmetric hyperbolic systems, exact controllability, observability, Carleman estimate

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1. Introduction and main results.

First-order hyperbolic systems govern the evolution of numerous physical quantities. Representative examples include the Saint–Venant equations for open channels, the Aw–Rascle equations for road traffic, gas dynamics, supply chains, and heat exchangers [2, 3, 11]. Due to their wide range of applications, the control theory of such systems has attracted considerable attention from both mathematical and engineering communities. Extensive research has been devoted to the boundary controllability of deterministic first-order hyperbolic systems (see, e.g., [6,10,12,17] and the references therein). Many classical results concern one-dimensional systems or multidimensional scalar equations; the present work treats a class of multidimensional systems whose principal matrices need not commute.

In many realistic scenarios, uncertainties–due to environmental noise, parameter fluctuations, or unmodeled dynamics–cannot be ignored. Such randomness is naturally incorporated through stochastic forcing, leading to systems of stochastic hyperbolic equations (e.g., [4, 5, 7, 16]). The presence of noise not only affects the well-posedness and stability but also fundamentally alters the controllability properties, as the control must counteract both deterministic and stochastic dynamics. A key question arises: Can we steer a stochastic multi-dimensional hyperbolic system from any initial state to any desired target state in finite time, and if so, under what conditions?

We address this question for symmetric systems under a matrix escape condition and a two-sided nested spectral boundary geometry. The boundary input is measured in the incoming flux norm. This distinction is essential near characteristic boundary points: ordinary unweighted boundary data cannot reach every terminal L^2 state even

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for scalar transport on a disk, as explained in subsection 2.5. Our weighted-energy approach is related to the Carleman method for deterministic symmetric hyperbolic systems in [8]; the stochastic proof must also account for the quadratic variation and the adapted control spaces. It yields a deterministic consequence under the same boundary hypotheses, without a claim of priority over the deterministic theory.

The symmetric principal matrices and the domain below are smooth. Their normal matrix may change rank under the stated nested geometry; no global eigenvector frame and no simultaneous diagonalization of the principal matrices are assumed. The threshold in the main theorem is a sufficient time, rather than a generally minimal time. Applications below verify the boundary geometry as well as the escape condition for genuinely noncommuting systems.

Let $G \subset \mathbb{R}^n$, $n \geq 2$, be a bounded smooth domain, with boundary Γ and outward unit normal ν. Throughout this paper, "smooth" means C^∞ up to the boundary. Set $H = L^2(G;\mathbb{R}^N)$, and let $A_j \in C^\infty(\overline{G};\mathbb{R}^{N\times N})$ be symmetric. Write

$$D = \sum_{j=1}^{n} A_j \partial_j, \qquad C = \sum_{j=1}^{n} \partial_j A_j, \qquad \mathsf{N} = \sum_{j=1}^{n} \nu_j A_j \quad \text{on } \Gamma. \tag{1.1}$$

The matrices A_i need not commute with A_j. For a symmetric matrix N define

$$\mathsf{P}_- = \mathbf{1}_{(-\infty,0)}(\mathsf{N}), \quad \mathsf{P}_+ = \mathbf{1}_{(0,\infty)}(\mathsf{N}), \quad \mathsf{N}_- = \tfrac{1}{2}(|\mathsf{N}| - \mathsf{N}), \quad \mathsf{N}_+ = \tfrac{1}{2}(|\mathsf{N}| + \mathsf{N}).$$

The projectors are measurable, including where ranks change. Let $E_{\geq 0}(\mathsf{N}) = \ker \mathsf{P}_-$ and $E_{\leq 0}(\mathsf{N}) = \ker \mathsf{P}_+$; these spaces include the kernel.

Assumption 1.1 (Two-sided nested spectral boundary geometry). *There is a finite disjoint union S of smooth closed embedded hypersurfaces of Γ (dimension $n-2$), possibly empty, such that* $\operatorname{Ran}\mathsf{N}$ *has constant dimension on each connected component of $\Gamma\setminus S$. Near each point of S, the following requirement holds for* each *of the boundary spaces $M = E_{\geq 0}(\mathsf{N})$ and $M = E_{\leq 0}(\mathsf{N})$: there exist smooth fixed-dimensional vector subbundles $M_{\mathrm{small}} \subseteq M_{\mathrm{big}}$ of the trivial boundary bundle on that entire neighborhood such that, on each side of S, M is one of these two bundles, and*

$$\ker \mathsf{N}(x) \subseteq M_{\mathrm{big}}(x) \tag{1.2}$$

throughout the neighborhood, in particular on S.

This is a condition on boundary matrices and smooth finite-dimensional bundles, not an assumption of operator generation, of a closed range, or of an adjoint domain identity. The requirement for both orientations is deliberate. The concrete criterion in Lemma A.1 makes it directly checkable. Away from S, constant rank and smoothness give smooth spectral subbundles. No globally smooth eigenframe, and no uniform inverse of a nonzero eigenvalue, is required. The smooth regularity in this theorem is a conservative choice; the minimal coefficient regularity is not investigated here.

Work on a complete probability space $(\Omega, \mathcal{F}, \mathbb{P})$ carrying a scalar Brownian motion W. Let $\mathbb{F} = (\mathcal{F}_t)_{0\leq t\leq T}$ be its augmented natural filtration. All process spaces below are progressively measurable. Put

$$\mathsf{S}^2(H) = L^2_{\mathbb{F}}(\Omega; C([0,T];H)), \qquad \mathsf{L}^2(H) = L^2_{\mathbb{F}}(0,T;H).$$

Assume

$$B_1, B_2, B_3 \in L^\infty_{\mathbb{F}}(0,T;L^\infty(G;\mathbb{R}^{N\times N})). \tag{1.3}$$

No spatial derivatives of these matrices are assumed.

The boundary Hilbert space with ordinary measure that parametrizes the input is

$$\mathcal{U} = \{h \in L^2(\Gamma;\mathbb{R}^N) : \mathsf{P}_- h = h \text{ a.e.}\}, \qquad \|h\|_{\mathcal{U}}^2 = \int_\Gamma |h|^2 \, d\Gamma. \tag{1.4}$$

It is a closed subspace. The physical incoming boundary datum can instead be written u in the equivalent flux space

$$U_- = \left\{ u : \ \mathsf{P}_- u = u, \quad \int_\Gamma u^\top \mathsf{N}_- u \, d\Gamma < \infty \right\}, \qquad h = \mathsf{N}_-^{1/2} u. \tag{1.5}$$

Measurable functions in U_- are identified a.e.; only the negative spectral subspace is included. The map in (1.5) is an isometric bijection onto $\mathcal{U}$: its inverse is the pointwise inverse on the negative spectral subspace, not a uniformly bounded matrix inverse.

Consider the equation

$$\begin{cases} dy + Dy\,dt = (B_1 y + B_3 v)\,dt + (B_2 y + v)\,dW_t & \text{in } G, \\ \mathsf{N}_-^{1/2} y = h & \text{on } \Gamma, \\ y(0) = y_0. \end{cases} \tag{1.6}$$

The boundary line is interpreted by the transposition identity (3.11), not as an assertion that every solution possesses an unweighted L^2 trace. When a classical trace exists, it is equivalent to $\mathsf{P}_- y = u$ under (1.5).

Assumption 1.2 (Escape multiplier). *There are $\eta \in C^1(\overline{G})$ and $c_0 > 0$ such that*

$$-\sum_{j=1}^{n} A_j(x)\partial_j \eta(x) \ \geq \ c_0 I_N \quad (x \in \overline{G}). \tag{1.7}$$

Write $d = \operatorname{osc}_{\overline{G}} \eta$ and $T_0 = d/c_0$.

Theorem 1.1 (Exact controllability). *Under Assumptions 1.1 and 1.2 and (1.3), for every $T > T_0$, $y_0 \in H$, and $y_1 \in L^2_{\mathcal{F}_T}(\Omega;H)$, there exist*

$$h \in L^2_{\mathbb{F}}(0,T;\mathcal{U}), \qquad v \in \mathsf{L}^2(H)$$

such that the unique transposition/mild solution of (1.6) belongs to $\mathsf{S}^2(H)$ and satisfies $y(T) = y_1$ almost surely. Equivalently, the boundary input belongs to $L^2_{\mathbb{F}}(0,T;U_-)$ in physical coordinates. There is a constant K_T, independent of y_0, y_1, for which

$$\|h\|_{L^2_{\mathbb{F}}(0,T;\mathcal{U})} + \|v\|_{\mathsf{L}^2(H)} \leq K_T\big(\|y_0\|_H + \|y_1\|_{L^2(\Omega;H)} \big).$$

The controls can be selected by a bounded linear map of (y_0, y_1).

The conclusion applies to all terminal L^2 random states and bounded random zeroth-order couplings in the stated flux-control space. The nested boundary geometry is part of the theorem; unrestricted variable-rank boundaries are not covered. No claim of optimality of T_0 is made.

The restrictions on control placement are stated and proved in Theorems 5.1 to 5.3. They concern the indicated additional hypotheses; they do not claim a universal minimal number of controls for every coupled system (see, for example, [1, 9]).

The remainder is organized as follows. Section 2 verifies concrete noncommuting examples and explains the flux norm and sufficient-time scope. Section 3 constructs the stochastic boundary trace and establishes forward well-posedness and duality. Section 4 proves observability and exact controllability, and Section 5 treats non-controllability. Appendix A establishes the deterministic operator realization used throughout, including both resolvent ranges and the exact adjoint domain.

2. Geometric examples and the scope of the theorem.

2.1. A noncommuting shallow-water model on the disk. Let $G = \{x_1^2 + x_2^2 < 1\}$ and

$$A_1 = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}, \qquad A_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}. \tag{2.1}$$

These matrices do not commute:

$$[A_1, A_2] = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} \neq 0.$$

Consequently there is no single orthogonal change of state variables that diagonalizes both principal matrices, even at one point. At $\nu = (\cos\theta, \sin\theta)$ the normal eigenvalues are

$$2\cos\theta - 1, \qquad 2\cos\theta, \qquad 2\cos\theta + 1.$$

They are separated from one another by 1. Their six zero points are distinct, and each crossing is simple since the corresponding $-2\sin\theta$ is nonzero. Local smooth spectral projections and Lemma A.1 verify Assumption 1.1. Take $\eta = -x_1$. Since $A_1 \geq I_3$, $c_0 = 1$ and $d = 2$. Therefore Theorem 1.1 gives exact controllability for every $T > 2$, with any bounded adapted B_1, B_2, B_3 of size 3×3 and every terminal state in $L^2_{\mathcal{F}_T}(\Omega; L^2(G; \mathbb{R}^3))$.

This is a genuine two-space-dimensional system. The diagonalization of its *normal* matrix along the boundary does not diagonalize the PDE. On this fixed disk the property persists under sufficiently small smooth symmetric C^1 perturbations of the two principal matrices: the boundary eigenvalues remain simple and separated, their six transverse zeros persist and remain disjoint, and $\widetilde{A}_1 \geq (1 - \varepsilon)I$. The sufficient time for these perturbed matrices is $T > 2/(1 - \varepsilon)$. Thus the example is not confined to constant coefficients.

2.2. An explicit variable-coefficient noncommuting family. Keep the disk and the two matrices of (2.1), but replace A_1 by

$$A_1^\varepsilon(x) = A_1 + \varepsilon x_2 I_3, \qquad 0 < |\varepsilon| \leq \tfrac{1}{4}.$$

Then $[A_1^\varepsilon(x), A_2] = [A_1, A_2] \neq 0$ at every point and $A_1^\varepsilon \geq (1 - |\varepsilon|)I_3$. The three normal eigenvalues are $a(\theta) - 1, a(\theta), a(\theta) + 1$, where $a(\theta) = \cos\theta(2 + \varepsilon\sin\theta)$. Their zero sets are disjoint. At a zero of $a \mp 1$, $|\cos\theta| \leq 4/7$, hence

$$|a'(\theta)| = |-2\sin\theta + \varepsilon(\cos^2\theta - \sin^2\theta)| \geq 2\sqrt{33}/7 - 1/4 > 0.$$

At a zero of a, $\cos\theta = 0$ and $|a'| \geq 2 - |\varepsilon| > 0$. All zeros are therefore transverse and, by compactness, finite. The normal spectral branches remain separated by 1,

so Lemma A.1 applies. Taking $\eta = -x_1$ proves exact controllability with arbitrary bounded random zeroth-order couplings for every $T > 2/(1-|\varepsilon|)$. In fact $C = 0$ for this family. This is a concrete variable-coefficient application, not only a constant-coefficient example or an appeal to generic perturbations.

2.3. An n-dimensional acoustic–convective class. Let $G = B_R(0) \subset \mathbb{R}^n$, $|U| > a > 0$, and define on $\mathbb{R}^{1+n}$

$$A_j = U_j I_{n+1} + a \begin{pmatrix} 0 & e_j^\top \\ e_j & 0_{n\times n} \end{pmatrix}. \tag{2.2}$$

Additional entropy/passive-transport components $U_j I_r$ may be adjoined. For different i, j, the commutator on the velocity block is $a^2(e_i e_j^\top - e_j e_i^\top)$ and is nonzero. The normal eigenvalues are

$$U\cdot\nu - a, \qquad U\cdot\nu + a, \qquad U\cdot\nu \quad \text{(the last with multiplicity } n-1+r).$$

Their zero sets on the sphere are three disjoint smooth latitude hypersurfaces, since $|U| > a$. The two acoustic projections on the first $1+n$ components are

$$\tfrac{1}{2}\begin{pmatrix} 1 \\ \nu \end{pmatrix}\begin{pmatrix} 1 & \nu^\top \end{pmatrix}, \qquad \tfrac{1}{2}\begin{pmatrix} 1 \\ -\nu \end{pmatrix}\begin{pmatrix} 1 & -\nu^\top \end{pmatrix}.$$

The convective projection is their orthogonal complement, plus the passive components. These projections are globally smooth; a globally smooth frame of the tangent bundle is neither asserted nor needed. Near the convective latitude, the whole multiple block crosses with the same sign. The other two blocks stay nonzero. Thus Lemma A.1 applies, including the multiple zero eigenvalue.

For $e = U/|U|$ and $\eta = -e\cdot x$,

$$-\sum_j A_j \partial_j \eta = \sum_j e_j A_j \geq (|U| - a) I, \qquad d = 2R.$$

Exact controllability follows for $T > 2R/(|U| - a)$. This includes the normalized constant-background acoustic–convective principal parts of linearized shallow-water and compressible-flow systems, not their full nonlinear dynamics. No statement is made that arbitrary perturbations of a multiple convective crossing preserve the nested geometry.

2.4. Symmetrizable systems with a positive time matrix.

COROLLARY 2.1. *Suppose the principal part is $A_0(x)dy + \sum_j A_j(x)\partial_j y\, dt$, where A_0 is smooth uniformly positive definite and A_j are smooth symmetric. Put $P = A_0^{1/2}$, $w = Py$, and $\widetilde{A}_j = P^{-1}A_jP^{-1}$. If the normalized matrices satisfy Assumptions* 1.1 *and* 1.2*, the flux-weighted exact controllability theorem holds for the symmetrizable system with full diffusion control, in the correspondingly transformed boundary coordinates.*

Proof. Multiply the equation by P^{-1} on the left and substitute $y = P^{-1}w$. The normalized time coefficient is I and the normalized spatial matrices are symmetric. Differentiating P^{-1} creates only bounded zeroth-order terms. For an additive diffusion input v, set $\widetilde{v} = P^{-1}v$; this is an isomorphism of the distributed control space. All transformed lower-order coefficients remain bounded and adapted. Apply Theorem 1.1 and transform back. Norms of initial and terminal states are equivalent by uniform positivity. In particular, the original matrices must not be inserted directly into an identity-time-matrix theorem without this normalization. ∎

2.5. Why the incoming flux norm is necessary. On the unit disk in coordinates (x, r) take $D = \partial_x$, zero lower-order coefficients, $y_0 = 0$, and $T > 2$. Set $s(r) = \sqrt{1 - r^2}$ and let $\Gamma^- = \{(-s(r), r) : -1 < r < 1\}$ be the inflow semicircle. At the left inflow point $(-s(r), r)$ one has $\mathsf{N}_- = s(r)$ and $d\Gamma = dr/s(r)$. The deterministic target $f(x, r) = (1 - r)^{-1/2}$ satisfies

$$\|f\|^2_{L^2(G)} = 2\int_{-1}^{1} \frac{\sqrt{1 - r^2}}{1 - r} dr = 2\pi.$$

Along the characteristic ending at (x, r), the mean incoming control must obey $\mathbb{E}u(T - x - s(r), -s(r), r) = f(x, r)$. Jensen's inequality then forces

$$\mathbb{E}\int_0^T \int_{\Gamma^-} |u|^2 d\Gamma dt \geq 2\int_{-1}^{1} \frac{dr}{1 - r} = \infty.$$

An internal diffusion control has zero mean and cannot remove this obstruction. The required input has finite flux norm instead: its norm squared on the used characteristics is $\int_G |f|^2 = 2\pi$. Thus merely repairing the old unweighted trace proof is impossible; the control norm really must change (or the target class must be reduced).

2.6. The time is sufficient, not generally minimal. For scalar $D = \partial_x$ on $G = \{(x, r) : (x - r)^2 + r^2 < 1\}$, the weight $\eta = -x$ gives $d = 2\sqrt{2}$, $c_0 = 1$. The alternative weight $\widetilde{\eta} = r - x$ has $d = 2$, $c_0 = 1$ on the same system and domain. Its scalar normal zeros are smooth simple glancing points, so the theorem already gives controllability for $T > 2$. Hence the threshold computed from an arbitrary admissible η is not generally a minimal time.

3. Well-posedness and boundary duality.

3.1. Backward stochastic equation and its flux trace. Let $\mathcal{A}$ be the graph closure of $-D$ with homogeneous incoming boundary condition, as defined in (A.3). By Proposition A.1, its adjoint is the graph realization of $D + C$ with homogeneous outgoing boundary condition, and both operators generate strongly continuous semigroups. The one-sided graph trace $\mathcal{C}_-$ is defined in Lemma A.3; below we extend it to backward stochastic solutions. Set $c = \|C\|_{L^\infty(G;\mathcal{L}(\mathbb{R}^N))}$ and denote the essential operator-norm bounds of B_j by b_j. Constants in the following well-posedness and trace estimates depend only on T, c, b_1, b_2, b_3.

For a terminal datum $z_T \in L^2_{\mathcal{F}_T}(\Omega; H)$ consider

$$dz = -\big(\mathcal{A}^* z + B_1^\top z + B_2^\top q\big)\, dt + q\, dW_t, \qquad z(T) = z_T. \tag{3.1}$$

Since $\mathcal{A}^*$ is a generator and the coefficients are bounded, the Hilbert-space backward stochastic evolution theorem under the natural Brownian filtration [13, Theorem 4.10] gives a unique mild solution $(z, q) \in \mathsf{S}^2(H) \times \mathsf{L}^2(H)$ and its continuous-dependence estimate. Set

$$Z = q + B_3^\top z. \tag{3.2}$$

Equivalently, the adjoint pair (z, Z) satisfies

$$dz + Dz\, dt = -\big[(C + B_1^\top - B_2^\top B_3^\top)z + B_2^\top Z\big]dt + (Z - B_3^\top z)\, dW_t.$$

It is Z, not q, that is dual to the distributed control v when $B_3 \neq 0$.

PROPOSITION 3.1 (Stochastic flux trace and weighted Itô identity). *There is a uniquely determined trace $\mathcal{C}_- z \in L^2_{\mathbb{F}}(0,T;\mathcal{U})$ obtained by graph-resolvent approximation, with*

$$\|\mathcal{C}_- z\|^2_{L^2_{\mathbb{F}}(0,T;\mathcal{U})} \le K \|z_T\|^2_{L^2(\Omega;H)}. \tag{3.3}$$

If $r \in C^1([0,T]\times\overline{G})$ is deterministic and positive, then

$$\begin{aligned}
&\mathbb{E}\int_G r(T)|z_T|^2 - \mathbb{E}\int_G r(0)|z(0)|^2 \\
&= \mathbb{E}\int_0^T\int_\Gamma r|\mathcal{C}_- z|^2 + \mathbb{E}\int_0^T\int_G z^\top \left(r_t I + \sum_j A_j \partial_j r - rC \right) z \\
&\quad - 2\mathbb{E}\int_0^T\int_G r\left\langle z, B_1^\top z + B_2^\top q\right\rangle + \mathbb{E}\int_0^T\int_G r|q|^2.
\end{aligned} \tag{3.4}$$

Space and time measures omitted from integral displays are $dx\,dt$ or $d\Gamma\,dt$ as appropriate. The same identity holds on every deterministic subinterval.

Proof. Let $R_\mu = \mu(\mu - \mathcal{A}^*)^{-1}$ for real $\mu > c$, and put

$$z_\mu = R_\mu z, \quad q_\mu = R_\mu q, \quad f = -B_1^\top z - B_2^\top q, \quad f_\mu = R_\mu f.$$

These satisfy the strong equation

$$dz_\mu = -\mathcal{A}^* z_\mu\,dt + f_\mu\,dt + q_\mu\,dW_t, \qquad z_\mu(T) = R_\mu z_T. \tag{3.5}$$

Indeed $R_\mu : H \to D(\mathcal{A}^*)$ and $\mathcal{A}^* R_\mu$ are bounded, so the assertion follows either from the mild equation and stochastic Fubini, or from the linear strong-solution criterion. In particular the noise is regularized too. Strong convergence and uniform boundedness of R_μ imply

$$z_\mu \to z \text{ in } \mathsf{S}^2(H), \qquad (q_\mu, f_\mu) \to (q,f) \text{ in } \mathsf{L}^2(H)^2.$$

For the first convergence, use compactness of each continuous sample path in H, followed by dominated convergence. We do *not* replace $R_\mu f$ by $-B_1^\top z_\mu - B_2^\top q_\mu$.

For $w = z_\mu - z_\rho$, $g = f_\mu - f_\rho$, and $p = q_\mu - q_\rho$, Itô's formula and (A.5) give

$$\begin{aligned}
\mathbb{E}\int_0^T \|\mathcal{C}_- w\|^2\,dt &= \mathbb{E}\,\|w(T)\|^2 - \mathbb{E}\,\|w(0)\|^2 + \mathbb{E}\int_0^T \langle Cw, w\rangle\,dt \\
&\quad - 2\mathbb{E}\int_0^T \langle w, g\rangle\,dt - \mathbb{E}\int_0^T \|p\|^2\,dt \\
&\le \mathbb{E}\,\|w(T)\|^2 + c\mathbb{E}\int_0^T \|w\|^2\,dt + 2\,\|w\|_{\mathsf{L}^2(H)}\,\|g\|_{\mathsf{L}^2(H)} \longrightarrow 0.
\end{aligned} \tag{3.6}$$

Thus the traces are Cauchy. Define $\mathcal{C}_- z$ to be their limit. A single-sequence version of this estimate, together with the backward solution estimate, gives (3.3). It also shows independence of the particular sequence of resolvent parameters. The limit is progressive and remains in $\mathcal{U}$.

Apply Hilbert-space Itô's formula to $\int_G r|z_\mu|^2$, using Lemma A.4 to integrate the unbounded operator. Pass to the limit by the preceding convergences, including the

boundary trace convergence. This gives (3.4). No differentiability of B_j is needed. To justify zero expectation of the local martingale, localization and BDG give

$$\mathbb{E}\left(\int_0^T |\langle rz_\mu, q_\mu\rangle_H|^2 dt\right)^{1/2} \le \|r\|_\infty \left(\mathbb{E}\sup_t \|z_\mu(t)\|^2\right)^{1/2} \left(\mathbb{E}\int_0^T \|q_\mu\|^2 dt\right)^{1/2} < \infty.$$

Thus it is an H^1 martingale. One must not infer square integrability of the product integrand from separate second moments; no fourth moments are imposed. ∎

3.2. Boundary admissibility and forward solutions. Let $S(t)$ be the semigroup generated by $\mathcal{A}$. Its adjoint is generated by $\mathcal{A}^*$. By (A.5), initially for $z \in D(\mathcal{A}^*)$,

$$\int_0^t \|\mathcal{C}_- S^*(r)z\|_{\mathcal{U}}^2 dr \le e^{ct}\|z\|_H^2. \tag{3.7}$$

Indeed the derivative of $e^{-cr}\|S^*(r)z\|^2$ is bounded above by $-e^{-cr}\|\mathcal{C}_- S^*(r)z\|^2$. Extend the observation by density in H. This is deterministic L^2 admissibility.

Let H_{-1} be the extrapolation space for $\mathcal{A}$, identified with the dual of $D(\mathcal{A}^*)$ using H as pivot. Define

$$\mathcal{K}: \mathcal{U} \to H_{-1}, \qquad \langle \mathcal{K}h, z\rangle_{H_{-1}, D(\mathcal{A}^*)} = \langle h, \mathcal{C}_- z\rangle_{\mathcal{U}}. \tag{3.8}$$

The deterministic boundary convolution, initially in H_{-1}, is

$$(\mathcal{G}h)(t) = \int_0^t S_{-1}(t-r)\mathcal{K}h(r)\, dr.$$

By duality with (3.7), it is H-valued and

$$\|(\mathcal{G}h)(t)\|_H \le e^{ct/2}\|h\|_{L^2(0,t;\mathcal{U})}, \qquad \sup_{t\le T}\|(\mathcal{G}h)(t)\|_H^2 \le e^{cT}\|h\|_{L^2(0,T;\mathcal{U})}^2. \tag{3.9}$$

It is continuous in H. To see this without a boundary lift, split the convolution at a fixed earlier time, use strong continuity of S on the old part, and use (3.9) to make the new short-time part small. For progressive random h, the same construction is causal, adapted, and belongs to $\mathsf{S}^2(H)$ by the pathwise bound.

Proposition 3.2 (Forward well-posedness). *For every $y_0 \in H$, $h \in L^2_{\mathbb{F}}(0,T;\mathcal{U})$ and $v \in \mathsf{L}^2(H)$ there is a unique $y \in \mathsf{S}^2(H)$ satisfying*

$$\begin{aligned} y(t) = S(t)y_0 + (\mathcal{G}h)(t) &+ \int_0^t S(t-r)(B_1 y + B_3 v)(r)\, dr \\ &+ \int_0^t S(t-r)(B_2 y + v)(r)\, dW_r, \end{aligned} \tag{3.10}$$

with the norm bounded by a constant times $\|y_0\| + \|h\|_{L^2_{\mathbb{F}}(0,T;\mathcal{U})} + \|v\|_{\mathsf{L}^2(H)}$.

Proof. The semigroup is quasi-contractive. Its stochastic convolution satisfies the usual Hilbert-space maximal inequality

$$\mathbb{E}\sup_{t\le\tau}\left\|\int_0^t S(t-r)g(r)\, dW_r\right\|_H^2 \le M_{c,\tau}\mathbb{E}\int_0^\tau \|g(r)\|_H^2 dr.$$

One obtains it by Itô's formula and BDG for Yosida approximations of the shifted dissipative generator, absorbing half the supremum and passing to the limit; the shift contributes only a bounded exponential factor. This is the quasi-contractive case of the stochastic evolution estimates in [13, Theorems 3.18(2) and 3.20]. The deterministic convolution estimate and this maximal inequality make the right-hand side of (3.10) a contraction on $\mathsf{S}^2(H)$ over a sufficiently short interval, since its Lipschitz square is bounded by $K(b_1^2\tau^2 + b_2^2\tau)$. The inhomogeneous term $\mathcal{G}h$ is in that space by (3.9). Iterate on finitely many intervals, using causality of the boundary convolution. The same estimates give uniqueness and continuous dependence on all data. ∎

PROPOSITION 3.3 (Transposition identity). *For y from Proposition* 3.2 *and (z, q) from* (3.1),

$$\mathbb{E}\,\langle y(T), z_T\rangle_H - \mathbb{E}\,\langle y_0, z(0)\rangle_H = \mathbb{E}\int_0^T \left(\langle h, \mathcal{C}_- z\rangle_{\mathcal{U}} + \langle v, Z\rangle_H\right) dt. \tag{3.11}$$

It holds at every terminal time $\tau \le T$ with the corresponding adjoint. Thus (3.10) *is the unique transposition solution of* (1.6).

Proof. Let $J_\mu = \mu(\mu - \mathcal{A})^{-1}$, and regularize only the unbounded boundary operator by $\mathcal{K}_\mu = J_{\mu,-1}\mathcal{K} : \mathcal{U} \to H$. In the regularized forward equation, $\mathcal{G}_\mu h = J_\mu \mathcal{G} h$. These boundary convolutions converge in $\mathsf{S}^2(H)$ by strong convergence of J_μ, compactness of continuous paths, and (3.9). Stability in Proposition 3.2 gives $y_\mu \to y$. For the now H-valued forcing, Hilbert-space forward–backward duality yields

$$\mathbb{E}\,\langle y_\mu(T), z_T\rangle - \mathbb{E}\,\langle y_0, z(0)\rangle = \mathbb{E}\int_0^T \left(\langle h, \mathcal{C}_- R_\mu z\rangle + \left\langle v, q + B_3^\top z\right\rangle\right) dt.$$

Here $J_\mu^* = R_\mu$ and $\mathcal{K}_\mu^* z = \mathcal{C}_- R_\mu z$. The Hilbert-space duality itself follows by applying resolvents to both mild solutions, applying the product rule to the resulting strong semimartingales, and passing to the limit in their already existing H-valued source and noise terms. The $\mathcal{A}, \mathcal{A}^*$ terms cancel exactly. One does not commute a resolvent through any B_j. The cross variation $\langle B_2 y + v, q\rangle$ cancels the B_2 adjoint drift, leaving $\left\langle v, q + B_3^\top z\right\rangle$. This also fixes the sign of the boundary term. Now use the trace convergence of Proposition 3.1 and $y_\mu \to y$ to obtain (3.11). The same argument works on $[0, \tau]$. Uniqueness in the transposition sense follows by testing against every $z_\tau \in L^2_{\mathcal{F}_\tau}(\Omega; H)$ at each time. ∎

Remark 3.1 (Relation to prescribed incoming data). The boundary operator in (3.8) is the physical flux operator, not an additional abstract actuator. For smooth y with incoming value u, Green's formula against $\mathsf{P}_+ z = 0$ contributes $-\int_\Gamma y^\top \mathsf{N} z = \int_\Gamma (\mathsf{N}_-^{1/2} u)\cdot \mathcal{C}_- z$. Smooth controls supported away from S and away from small negative eigenvalues are dense in $\mathcal{U}$: first cut off those sets in L^2, then use smooth local spectral projectors and a boundary partition of unity. When a zero block persists off S, it is simply not controlled. Hence the completed input model agrees with the finite-flux completion of ordinary smooth incoming inputs; it does not impose an additional unmentioned compatibility condition on h.

4. Observability and exact controllability. Using the bounds c, b_j fixed in Section 3, define

$$K_0 = c + 2b_1 + b_2^2, \qquad K = c + 2b_1 + 2(b_2 + b_3)^2. \tag{4.1}$$

LEMMA 4.1 (Endpoint energy comparison). *For $0 \le t_1 \le t_2 \le T$,*

$$\mathbb{E}\,\|z(t_1)\|_H^2 \le e^{K_0(t_2-t_1)}\mathbb{E}\,\|z(t_2)\|_H^2\,. \tag{4.2}$$

Proof. Use (3.4) with $r = 1$ on $[t_1, t_2]$. Complete the square:

$$|q|^2 - 2\,\langle B_2 z, q\rangle = |q - B_2 z|^2 - |B_2 z|^2 \ge -b_2^2|z|^2.$$

The nonnegative boundary term can be dropped. Thus the absolutely continuous function $e(t) = \mathbb{E}\,\|z(t)\|_H^2$ obeys $e'(t) \ge -K_0 e(t)$ a.e. Integrating the differential inequality proves the result. ∎

PROPOSITION 4.1 (Observability). *Let $T > d/c_0$. Choose*

$$\begin{aligned} &\beta = \frac{1}{2}\left(c_0 + \frac{d}{T}\right), \qquad \delta = c_0 - \beta = \frac{1}{2}\left(c_0 - \frac{d}{T}\right) > 0,\\ &s \ge \max\left\{\frac{K+1}{2\delta}, \frac{K_0 T + \log 2}{2\delta T}\right\}. \end{aligned} \tag{4.3}$$

Then

$$\mathbb{E}\,\|z_T\|_H^2 \le C_{\rm obs}\mathbb{E}\int_0^T \big(\,\|\mathcal{C}_- z\|_{\mathcal{U}}^2 + \|Z\|_H^2\,\big)dt, \qquad C_{\rm obs} = 4e^{2sd}. \tag{4.4}$$

Proof. Add a constant to η so that $\min\eta = 0$ and $\max\eta = d$, and set $r = e^{2s(\eta+\beta t)}$. By (1.7),

$$r_t I + \sum_j A_j\partial_j r = 2sr\left(\beta I + \sum_j A_j\partial_j\eta\right) \le -2s\delta r I.$$

Using $q = Z - B_3^\top z$, the remaining noise contribution satisfies

$$\begin{aligned} |q|^2 - 2\,\langle B_2 z, q\rangle &= |Z - (B_3^\top + B_2)z|^2 - |B_2 z|^2\\ &\le 2|Z|^2 + 2(b_2 + b_3)^2|z|^2. \end{aligned} \tag{4.5}$$

The other zeroth-order terms are bounded above by $(c + 2b_1)r|z|^2$. Therefore (3.4) yields

$$\begin{aligned} &\mathbb{E}\int_G r(T)|z_T|^2 + (2s\delta - K)\mathbb{E}\int_0^T\int_G r|z|^2\\ &\qquad \le \mathbb{E}\int_G r(0)|z(0)|^2 + \mathbb{E}\int_0^T\int_\Gamma r|\mathcal{C}_- z|^2 + 2\mathbb{E}\int_0^T\int_G r|Z|^2. \end{aligned} \tag{4.6}$$

All identities have already been justified for the actual mild solution, rather than merely for a formal differentiable solution.

Since $\beta T - d = \delta T$, Lemma 4.1 and (4.3) give

$$\mathbb{E}\int_G r(0)|z(0)|^2 \le e^{2sd + K_0 T}\mathbb{E}\,\|z_T\|^2 \le \tfrac{1}{2}e^{2s\beta T}\mathbb{E}\,\|z_T\|^2 \le \tfrac{1}{2}\mathbb{E}\int_G r(T)|z_T|^2.$$

Also $2s\delta - K \ge 1$, so the interior term can be dropped. Finally $r(T, x) \ge e^{2s\beta T}$ and $r(t, x) \le e^{2s(\beta T + d)}$. Divide (4.6) by $\frac{1}{2}e^{2s\beta T}$ to obtain (4.4). No commutator of two principal matrices appears. ∎

4.1. Completion of the controllability proof. Let $\mathcal{X} = L^2_{\mathcal{F}_T}(\Omega; H)$ and $\mathcal{V} = L^2_{\mathbb{F}}(0,T;\mathcal{U}) \times \mathsf{L}^2(H)$, both real Hilbert spaces. Define

$$\mathcal{O} : \mathcal{X} \to \mathcal{V}, \qquad \mathcal{O} z_T = (\mathcal{C}_- z, Z).$$

The backward estimates and (3.3) show boundedness, and (4.4) shows that $\mathcal{O}$ is injective with closed range. For fixed y_0, y_1, define on that range

$$\ell(\mathcal{O} z_T) = \mathbb{E}\,\langle y_1, z_T\rangle - \mathbb{E}\,\langle y_0, z(0)\rangle\,.$$

It is well-defined, and by (4.2),

$$|\ell(\mathcal{O} z_T)| \le \sqrt{C_{\mathrm{obs}}}\big(\,\|y_1\|_{\mathcal{X}} + e^{K_0 T/2}\,\|y_0\|_H\,\big)\,\|\mathcal{O} z_T\|_{\mathcal{V}}\,. \tag{4.7}$$

Riesz representation on the closed Hilbert subspace $\operatorname{Ran}\mathcal{O}$ gives a unique $(h, v) \in \operatorname{Ran}\mathcal{O} \subseteq \mathcal{V}$ representing ℓ. These controls are progressive because the ambient space already imposes progressiveness. Their product norm is bounded by the prefactor in (4.7). The representing vector depends linearly and continuously on (y_0, y_1). Using (3.11) shows

$$\mathbb{E}\,\langle y(T) - y_1, z_T\rangle = 0 \quad \text{for all } z_T \in \mathcal{X},$$

hence $y(T) = y_1$ in $\mathcal{X}$ and almost surely. This proves Theorem 1.1. The constructed pair is also the minimum-product-norm control: any two controls giving the same terminal state differ by a vector in $\ker\mathcal{O}^* = (\operatorname{Ran}\mathcal{O})^{\perp}$.

Corollary 4.1 (Deterministic boundary-only consequence). *Under the same principal and boundary assumptions, the deterministic equation $y_t + Dy = B_1(t,x)y$, with bounded deterministic B_1, is exactly controllable between arbitrary states of H by the flux-weighted incoming boundary control alone whenever $T > T_0$.*

Proof. For its deterministic adjoint set $q = Z = 0$ in the weighted identity and the observability argument. The resulting inequality observes only $\mathcal{C}_- z$. Deterministic admissibility and Hilbert-space duality give the assertion. This is a consequence of the argument; no priority claim relative to the existing deterministic literature is made. □

5. Lack of exact controllability. Here Assumption 1.1 is sufficient; the escape multiplier is not needed. We use compactly supported tests, so the boundary term vanishes also for finite-flux controls.

Lemma 5.1 (A terminal random variable with no continuous-noise representation). *Let*

$$g(t) = \sin\!\left(\frac{1}{T-t}\right) \quad (t < T), \qquad \rho = \int_0^T g(t)\,dW_t.$$

There do not exist $a \in L^2_{\mathbb{F}}(0,T;\mathbb{R})$, a progressive $b \in C([0,T];L^2(\Omega))$, and a deterministic constant a_0 such that

$$\rho = a_0 + \int_0^T a(t)\,dt + \int_0^T b(t)\,dW_t. \tag{5.1}$$

Proof. If such a representation existed, subtract its conditional expectation at time t. Adaptedness, martingale increments and Itô's isometry imply

$$\mathbb{E}\int_t^T |g(s) - b(s)|^2 ds = \mathbb{E}\left|\int_t^T a(s)ds - \mathbb{E}\left(\int_t^T a(s)ds \mid \mathcal{F}_t\right)\right|^2$$

$$\leq (T-t)\mathbb{E}\int_t^T |a(s)|^2 ds.$$

After division by $T-t$ the right-hand side tends to zero. Continuity of b at T in $L^2(\Omega)$ permits replacing $b(s)$ by $b(T)$ in this normalized mean-square integral. Integration by parts after the change of variables $r = (T-s)^{-1}$ gives

$$\frac{1}{T-t}\int_t^T g(s)ds \longrightarrow 0, \qquad \frac{1}{T-t}\int_t^T g(s)^2 ds \longrightarrow \tfrac{1}{2}.$$

Consequently the normalized integral with $b(T)$ tends to $\frac{1}{2}+\mathbb{E}|b(T)|^2$, a contradiction. This is an explicit proof of the type of obstruction used in [13, Proposition 6.2]; no novelty claim is made for the obstruction mechanism. ∎

For $\chi \in C_c^\infty(G;\mathbb{R}^N)$, the forward solution satisfies

$$\begin{aligned}\langle y(t),\chi\rangle_H = \langle y_0,\chi\rangle_H &+ \int_0^t \big[\,\langle y, \mathcal{A}^*\chi + B_1^\top\chi\rangle_H + \langle v, B_3^\top\chi\rangle_H\,\big]ds \\ &+ \int_0^t \big[\,\langle y, B_2^\top\chi\rangle_H + \langle v,\chi\rangle_H\,\big]dW_s. \end{aligned} \tag{5.2}$$

Indeed $\chi \in D(\mathcal{A}^*)$ and $\mathcal{C}_-\chi = 0$. Apply the weak equation to the bounded-boundary-operator approximations from Proposition 3.3 and pass to the limit: $\mathcal{C}_- R_\mu\chi \to \mathcal{C}_-\chi = 0$ in the graph trace norm. The boundary term therefore disappears even for the enlarged flux-control space.

THEOREM 5.1. *Suppose B_1 is deterministic and $B_3 = 0$. With $h = 0$, system (1.6) is not exactly controllable for any $T > 0$.*

THEOREM 5.2. *Suppose $B_2 \in C([0,T];L^\infty(\Omega\times G;\mathbb{R}^{N\times N}))$ with adapted values. Let G_0 be an open subset of G with $G\setminus\overline{G_0}\neq\varnothing$. If v is supported in $(0,T)\times G_0$, system (1.6) is not exactly controllable for any $T>0$, even with arbitrary incoming flux control.*

THEOREM 5.3. *Under the same time-continuity assumption on B_2 as in Theorem 5.2, the system with only an internal drift control,*

$$dy + Dy\,dt = (B_1 y + w)\,dt + B_2 y\,dW_t, \qquad \mathsf{N}_-^{1/2} y = h, \qquad y(0) = y_0,$$

is not exactly controllable for any $T>0$, where $w \in \mathsf{L}^2(H)$ and $h \in L^2_{\mathbb{F}}(0,T;\mathcal{U})$ are arbitrary.

Proof. For Theorem 5.1, take $y_0 = 0$. The mean $\mathbb{E}y$ solves the deterministic homogeneous mild equation with generator $\mathcal{A}$ and bounded deterministic coefficient B_1. Uniqueness gives $\mathbb{E}y(t) = 0$. A nonzero deterministic terminal state is impossible.

For Theorem 5.2, choose a unit vector-valued test function $\chi \in C_c^\infty(G\setminus\overline{G_0};\mathbb{R}^N)$ and use the target $y_1 = \rho\chi$ from Lemma 5.1. In (5.2), both terms involving v vanish by spatial support, because B_3 is a multiplication operator. The drift integrand is in $L^2_{\mathbb{F}}(0,T)$ and the noise integrand $b(t) = \langle B_2(t)y(t),\chi\rangle$ is continuous in $L^2(\Omega)$: use continuity of y in $L^2(\Omega;H)$, uniform boundedness of B_2, and its asserted L^∞-norm time continuity. At time T one obtains the representation (5.1), a contradiction. For Theorem 5.3, choose any unit $\chi \in C_c^\infty(G;\mathbb{R}^N)$. The added drift term $\langle w,\chi\rangle$ is in L^2, and the noise integrand is unchanged. The same contradiction applies. ∎

Appendix A. Boundary geometry and the operator realization.

A.1. The boundary criterion and the classical weak–strong principle.

LEMMA A.1 (A directly verifiable coherent-crossing class). *Suppose* $\operatorname{Ran}\mathsf{N}$ *has constant dimension away from a finite disjoint union* S *as above. Near every point of* S*, suppose a smooth local orthogonal frame gives*

$$Q^{\top}\mathsf{N}Q = \operatorname{diag}(D_p, \kappa D_c, 0, -D_m), \tag{A.1}$$

where the nonempty D_p, D_c, D_m *are smooth positive definite blocks. There is a smooth local defining function* h_0 *of* S *with* $d_\Gamma h_0 \neq 0$*, such that* $\kappa = 0$ *on* S *and* $\kappa h_0 > 0$ *off* S*. Then Assumption* 1.1 *holds. The zero block and either of* D_p, D_m *may be absent.*

Proof. Let the smooth bundles associated with the four blocks be E_p, E_c, E_0, E_m. For N, the small and big allowed bundles are $E_p \oplus E_0$ and $E_p \oplus E_c \oplus E_0$. For $-\mathsf{N}$ they are $E_m \oplus E_0$ and $E_m \oplus E_c \oplus E_0$, with the sides interchanged. At S, $\ker\mathsf{N} = E_c \oplus E_0$, which belongs to both corresponding big bundles. All required inclusions and side identifications follow. ∎

Only the normal matrix is put into a *local boundary* block form. No simultaneous diagonalization of $A_1, \dots, A_n$ in the domain occurs. The common special case is $\kappa = h_0$. Even $\kappa = h_0^{2k+1}$ is allowed: the proof needs the bundle geometry, not a nonzero derivative of each eigenvalue. Oppositely directed zero crossings at the same interface need not satisfy the nested condition and are not automatically included.

LEMMA A.2 (Imported weak–strong principle). *For a smooth symmetric first-order operator* $L = \sum_j a_j\partial_j + b$ *on a smooth bounded domain, equip the boundary with a maximal nonnegative space* M *for* a_ν*. Assume the rank is constant off smooth embedded boundary interfaces and* M *admits the local nested small/big extensions containing* $\ker a_\nu$ *as in Assumption* 1.1*. Every* L^2 *weak solution of* $(\lambda + L)w = f$*, with this homogeneous boundary condition, is strong in the graph sense: there exist smooth* w_k *satisfying that condition off the interfaces with*

$$w_k \longrightarrow w, \qquad (\lambda + L)w_k \longrightarrow f \quad \text{in } L^2.$$

Here weak means testing against smooth functions in the adjoint boundary space $M^* = (a_\nu M)^{\perp}$*.*

Source and meaning. We use the classical weak–strong theorem of Rauch [15] in the form explicitly stated in Nishitani–Takayama [14, Section 1, pp. 168–169] and Secchi [19, Section 1]. The weak formulation is also explicitly given in [14, p. 170]. The graph approximation with exact homogeneous boundary constraints is illustrated in [18, Proposition 1.3 and its proof], which cites Rauch's Corollary 2.5. That proposition's *special* sign-changing class is not being mistaken for the general mixed-sign theorem. No H^1-regularity conclusion, flow-sign condition for higher regularity, or small-coupling condition is imported. This classical result is the one non-elementary deterministic boundary input. We invoke the nested-bundle formulations stated in these references; we do not assert weak–strong equivalence for arbitrary variable-rank boundary geometry.

For our N, $E_{\geq 0}(\mathsf{N})$ is maximal nonnegative and

$$(\mathsf{N}E_{\geq 0}(\mathsf{N}))^{\perp} = E_{\leq 0}(\mathsf{N}). \tag{A.2}$$

Indeed, $\mathsf{N}E_{\geq 0}(\mathsf{N}) = E_{>0}(\mathsf{N})$. The reversed identity holds for $-\mathsf{N}$. Thus Lemma A.2 applies in both orientations under Assumption 1.1, including the noncommuting examples below.

A.2. Both resolvent ranges and the exact adjoint domain. Define the smooth cores

$$\mathscr{D}_+ = \{y \in C^\infty(\overline{G};\mathbb{R}^N) : \mathsf{P}_-\gamma y = 0\}, \qquad \mathscr{D}_- = \{z \in C^\infty(\overline{G};\mathbb{R}^N) : \mathsf{P}_+\gamma z = 0\},$$

where constraints can equivalently be imposed off S. Define the graph closures

$$\mathcal{A} = \overline{(-D, \mathscr{D}_+)}, \qquad \mathcal{B} = \overline{(D + C, \mathscr{D}_-)}. \tag{A.3}$$

These differential operators are closable, by distributional convergence, and densely defined because their cores contain $C_c^\infty(G;\mathbb{R}^N)$. Let $c = \|C\|_{L^\infty(G;\mathcal{L}(\mathbb{R}^N))}$.

Lemma A.3 (One-sided flux traces). *There are continuous graph-norm maps*

$$\mathcal{C}_+ : D(\mathcal{A}) \to L^2(\Gamma;\mathbb{R}^N), \quad \mathcal{C}_- : D(\mathcal{B}) \to \mathcal{U}$$

which on the respective smooth cores are $\mathsf{N}_+^{1/2}\gamma y$ *and* $\mathsf{N}_-^{1/2}\gamma z$. *They satisfy*

$$2\langle \mathcal{A}y, y\rangle_H = -\|\mathcal{C}_+ y\|^2_{L^2(\Gamma)} + \langle Cy, y\rangle_H, \tag{A.4}$$

$$2\langle \mathcal{B}z, z\rangle_H = -\|\mathcal{C}_- z\|^2_{\mathcal{U}} + \langle Cz, z\rangle_H. \tag{A.5}$$

Furthermore $\mathcal{B} \subseteq \mathcal{A}^*$ *and* $\mathcal{A} \subseteq \mathcal{B}^*$.

Proof. Integration by parts on the cores gives the two energy identities. For example,

$$\left\|\mathsf{N}_-^{1/2}\gamma z\right\|^2_{\mathcal{U}} \le c\|z\|^2_H + 2\|(D+C)z\|_H \|z\|_H.$$

Apply this to differences of a graph-approximating sequence. The flux traces are Cauchy in the indicated Hilbert spaces, independent of the sequence, and the identities pass to the limit. On the smooth cores the cross-boundary form $\int_\Gamma y^\top \mathsf{N} z$ vanishes, since y is in the nonnegative space and z in the nonpositive space. The cross Green identity therefore passes to the graph closures and gives the two inclusions. This argument asserts no trace theorem on the unrestricted maximal graph space. ∎

Proposition A.1 (Range and adjoint theorem). *Under Assumption* 1.1, *for every real* $\lambda > c/2$,

$$\mathrm{Ran}(\lambda - \mathcal{A}) = H, \qquad \mathrm{Ran}(\lambda - \mathcal{B}) = H, \qquad \left\|(\lambda-\mathcal{A})^{-1}\right\|, \left\|(\lambda-\mathcal{B})^{-1}\right\| \le (\lambda - c/2)^{-1}. \tag{A.6}$$

Consequently

$$\mathcal{A}^* = \mathcal{B}, \qquad \mathcal{B}^* = \mathcal{A}, \tag{A.7}$$

and both generate strongly continuous semigroups of norm at most $e^{ct/2}$.

Proof. We prove the first range statement without assuming any identity of adjoint domains. Given $f \in H$, for $v \in \mathscr{D}_-$, (A.5) implies

$$\|(\lambda - D - C)v\|_H \ge (\lambda - c/2)\|v\|_H.$$

The functional on $(\lambda - D - C)\mathscr{D}_-$ defined by

$$(\lambda - D - C)v \longmapsto \langle f, v\rangle_H$$

is well-defined and has norm at most $(\lambda - c/2)^{-1} \|f\|_H$. Hahn–Banach and Riesz give $u \in H$ such that

$$\langle u, (\lambda - D - C)v \rangle_H = \langle f, v \rangle_H \quad (v \in \mathscr{D}_-). \tag{A.8}$$

By (A.2), this is exactly the weak formulation of $(\lambda + D)u = f$ with boundary space $E_{\geq 0}(\mathsf{N})$. The shifted operator is symmetric positive because $2\lambda I - C \geq (2\lambda - c)I$. The hypotheses of Lemma A.2 hold by Assumption 1.1. Hence there are $u_k \in \mathscr{D}_+$ with $u_k \to u$ and $(\lambda + D)u_k \to f$ in H. Thus $u \in D(\mathcal{A})$ and $(\lambda - \mathcal{A})u = f$. The energy inequality gives injectivity and the stated inverse bound.

For the second range statement apply the identical construction to $L = -D - C$, whose normal matrix is $-\mathsf{N}$, with maximal nonnegative space $E_{\leq 0}(\mathsf{N})$. Its formal adjoint is D and its shifted symmetric part is again $2\lambda I - C$. The second orientation in Assumption 1.1 is exactly what is required here. This proves both parts of (A.6).

The energy identities show that $\mathcal{A} - cI/2$ and $\mathcal{B} - cI/2$ are dissipative; (A.6) and the Lumer–Phillips theorem give the semigroups. For completeness, let $z \in D(\mathcal{A}^*)$, and choose $w \in D(\mathcal{B})$ with $(\lambda - \mathcal{B})w = (\lambda - \mathcal{A}^*)z$. The inclusion in Lemma A.3 yields $(\lambda - \mathcal{A}^*)(z - w) = 0$. Since $\mathrm{Ran}(\lambda - \mathcal{A}) = H$, the kernel of $\lambda - \mathcal{A}^*$ is zero, so $z = w$. Thus $D(\mathcal{A}^*) \subseteq D(\mathcal{B})$, proving the first equality in (A.7); taking adjoints proves the second. No energy estimate was applied to an unrecognized weak adjoint domain. ∎

Lemma A.4 (Scalar multiplier invariance). *For $a \in C^1(\overline{G})$, multiplication by a preserves $D(\mathcal{A})$ and $D(\mathcal{A}^*)$. In particular,*

$$\mathcal{A}^*(az) = a\mathcal{A}^* z + \sum_j A_j(\partial_j a)z, \qquad \mathcal{C}_-(az) = a|_\Gamma \, \mathcal{C}_- z.$$

For real a,

$$2\langle az, \mathcal{A}^* z \rangle_H = -\int_\Gamma a|\mathcal{C}_- z|^2 \, d\Gamma + \int_G z^\top \left(aC - \sum_j A_j \partial_j a \right) z \, dx. \tag{A.9}$$

Proof. For smooth scalar a, multiplication preserves the boundary cores, and the commutator is a bounded zeroth-order operator. Graph approximation proves the claim for domain elements. Approximate a C^1 scalar by smooth scalars in the C^1 norm to obtain the stated regularity. Weighted integration by parts on the core followed by graph convergence proves (A.9). Scalar multiplication is essential: no corresponding assertion for arbitrary smooth matrix multipliers on these graph spaces is used. ∎

A.3. No universal graph-space trace and no global frame. On a flat boundary patch with inward coordinate $r > 0$ and tangential coordinates s, q, take

$$A_s = I_2, \quad A_r = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad A_q = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The functions $e^{-kr}(\cos kq, \sin kq)^\top$ satisfy $(A_r\partial_r + A_q\partial_q)u = 0$. Multiply by fixed smooth cutoffs, one equal to one near $r = 0$, and by $\sqrt{k}$. Their L^2 graph norms remain bounded, whereas their boundary L^2 norms grow like $\sqrt{k}$. This invalidates an unrestricted graph-norm-to-L^2 trace assertion, not Lemma A.3, whose domains impose one-sided homogeneous boundary conditions. The proof also never assumes a global smooth eigenvector frame of N.

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